2026-06-05 DNA Quantification — Portelite BDR (Qubit green channel)

Processes the Qubit fluorescence data recorded in `../lab_notebook/digitalized/2026-06-05_plaque-assay-r3-dna-extraction-quantification.md` (source page p.31).

Kit: Portelite™ Fluorimetric DNA Quantitation Kit, Broad Dynamic Range (AAT Bioquest 17665). Ex/Em 480/530 nm, green channel. Standards are calf-thymus DNA; the curve is fit by **linear regression of RFU vs. concentration (ng/µL)** after subtracting the Std-0 blank RFU.

Three sections:

1. Build the standard curve (linear fit → slope, intercept, R^2).
2. Convert each sample's reading to a DNA concentration (ng/µL).
3. Compute the DNA mass currently in each fridge tube (concentration × remaining volume).

Key data note (read before trusting any number)

The sample columns in the .md are headed "Reading (ng/µL)" but the values are actually the **raw green-channel fluorescence (RFU)**, on the *same scale as the standards* — e.g. Std 100 reads 16695 RFU while sample 8-2-2 reads 43887. (On p.31 the team headed the whole block "Absorbance @540 nm".) Section 1 builds the curve; Section 2 turns these RFU into concentrations.

Assumptions (flagged — change at the top if wrong)

- `DILUTION_FACTOR = 1` — no record of pre-diluting samples; the 10 µL taken into the assay is the same volume the standards use, and ~90/80 µL "left after aliquoting" implies a ~100 µL eluate drawn undiluted. If any sample was diluted before reading, set this factor accordingly.
- **Remaining volumes (from p.31 margin):** bacterial-genome extracts (Set A) ~**90 µL**; metagenomic extracts and mixes (Set B) ~**80 µL**. All stored at **4 °C**.
- Sample IDs / readings are transcribed verbatim from the .md (incl. `8–3–1` in Set A and the mix labels M1/M3/M4/M7 from the data table).

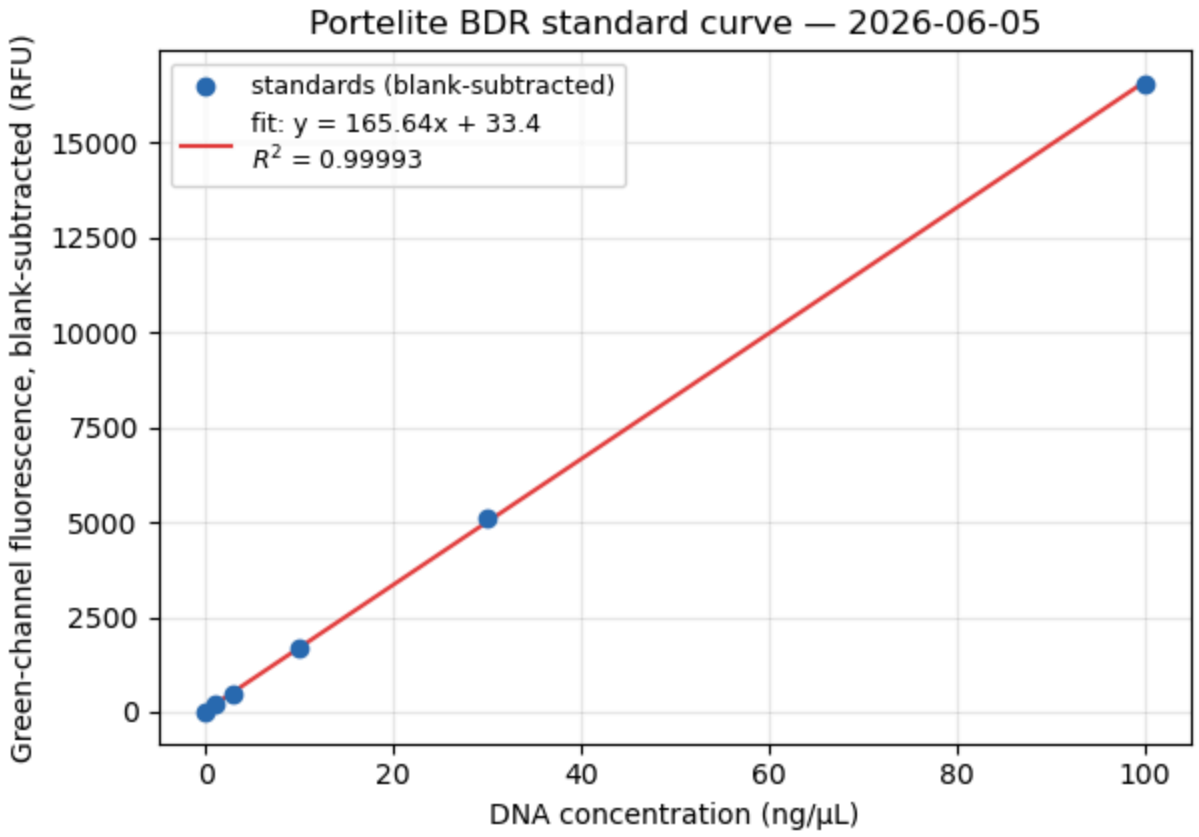
Section 1 — Build the standard curve

Six-point series (the team used Component D = 100 ng/µL directly as the top standard). Subtract the Std-0 blank RFU from every reading, then fit `RFU_corrected = slope · conc +`

`intercept` by linear regression. Report slope, intercept and R².

```
Blank (Std 0) RFU      : 127.59
Slope                  : 165.6425 RFU per ng/uL
Intercept              : 33.4265 RFU
R^2                    : 0.999929
Top standard           : 100 ng/uL (readings above this are extrapolated)
```

	standard	conc_ng_uL	rfu	rfu_corrected
0	Std 0 (blank)	0.0	127.59	0.00
1	Std 1	1.0	325.63	198.04
2	Std 3	3.0	606.82	479.23
3	Std 10	10.0	1833.66	1706.07
4	Std 30	30.0	5229.49	5101.90
5	Std 100	100.0	16695.43	16567.84



Section 2 — Sample DNA concentrations

Convert each raw reading to ng/µL via the fitted curve. `extrapolated = True` flags samples whose concentration lands above the 100 ng/µL top standard (per the PI note, these are quoted off the extrapolated line as long as they are not orders of magnitude above 100).

Set A — bacteria-genome extracts

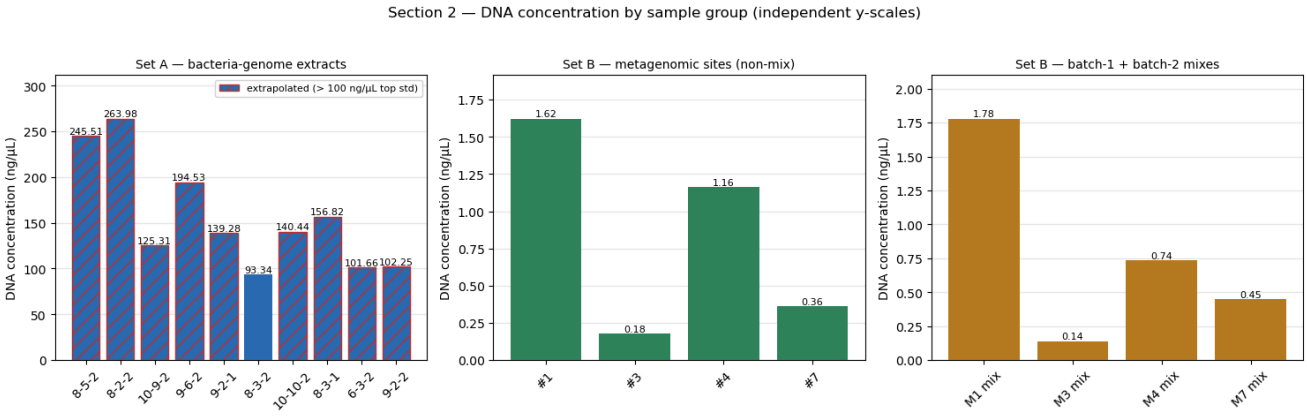
sample	rfu	conc_ng_uL	extrapolated
8-5-2	40,827.88	245.51	True
8-2-2	43,887.14	263.98	True
10-9-2	20,917.70	125.31	True
9-6-2	32,384.09	194.53	True
9-2-1	23,232.52	139.28	True
8-3-2	15,622.18	93.34	False
10-10-2	23,423.31	140.44	True
8-3-1	26,136.76	156.82	True
6-3-2	16,999.92	101.66	True
9-2-2	17,097.18	102.25	True

Set B — metagenomic sites + mixes

sample	rfu	conc_ng_uL	extrapolated
#1	429.84	1.62	False
#3	190.14	0.18	False
#4	353.60	1.16	False
#7	220.63	0.36	False
M1 mix	455.83	1.78	False
M3 mix	184.11	0.14	False
M4 mix	283.19	0.74	False
M7 mix	235.20	0.45	False

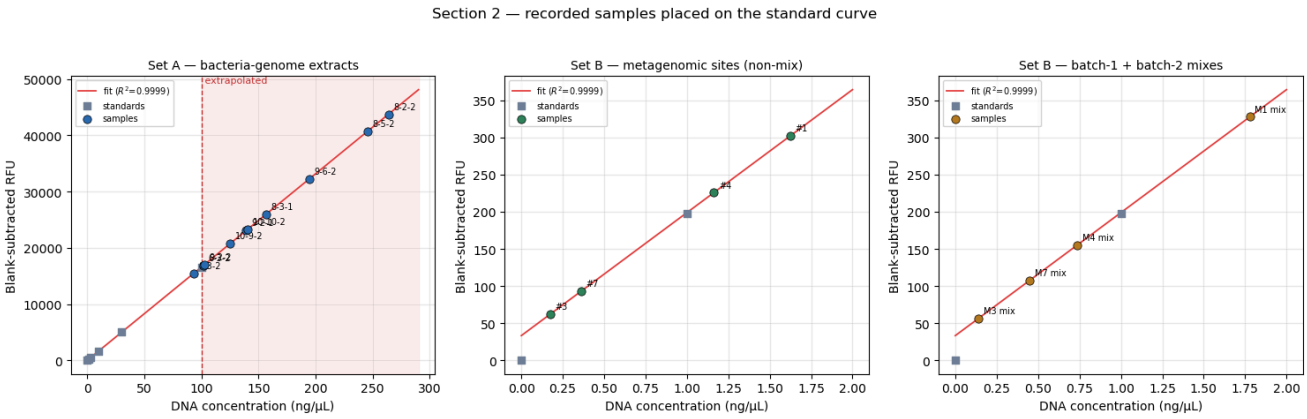
Section 2 plots — concentration by sample group

Three bar charts with **independent y-axes** (bacterial reads ~100× higher than metagenomic, so a shared scale would flatten Set B to invisibility): bacteria-genome extracts, non-mix metagenomic sites, and batch-1 + batch-2 mixes. Hatched red-edged bars are **extrapolated** above the 100 ng/µL top standard.



Section 2 plots — recorded samples on the standard curve

The same three groups, but here each sample is placed **on the standard curve** at its (concentration, blank-subtracted RFU) position, alongside the standards (grey squares) and the fitted line. Per-group x/y ranges differ widely, so each panel is auto-scaled. For Set A the shaded band marks the **extrapolated region beyond the 100 ng/µL top standard**.



Section 3 — DNA mass currently in each fridge tube

`mass = concentration (ng/µL) × remaining volume (µL)`, reported in µg. Remaining volume is the post-aliquot "left in tube" amount from the p.31 margin: **90 µL** for Set A (bacterial), **80 µL** for Set B (metagenomic sites + mixes). All tubes are at **4 °C**.

> Assumption flag: the 80 µL margin note is written for "metagenomic samples" — applied here to the mixes as well, since they are metagenomic. Adjust `VOL_METAGENOMIC_uL` per-mix if they differ.

```
Set A — bacteria-genome extracts (90 uL each, 4 C)
 sample  conc_ng_uL  remaining_vol_uL  mass_ug  extrapolated
  8-5-2     245.510            90.000   22.096          True
  8-2-2     263.979            90.000   23.758          True
 10-9-2     125.310            90.000   11.278          True
  9-6-2     194.534            90.000   17.508          True
  9-2-1     139.285            90.000   12.536          True
  8-3-2      93.341            90.000    8.401         False
10-10-2     140.437            90.000   12.639          True
  8-3-1     156.818            90.000   14.114          True
  6-3-2     101.658            90.000    9.149          True
  9-2-2     102.245            90.000    9.202          True

Set B — batch-1 + batch-2 mixes (80 uL each, 4 C)
sample  conc_ng_uL  remaining_vol_uL  mass_ug  extrapolated
M1 mix       1.780            80.000    0.142         False
M3 mix       0.139            80.000    0.011         False
M4 mix       0.738            80.000    0.059         False
M7 mix       0.448            80.000    0.036         False

Total recoverable DNA — Set A (bacterial): 140.68 ug
Total recoverable DNA — Set B mixes: 0.248 ug
```

Readout. Set A bacterial genomes are in the ~90–260 ng/µL range (≈ 8–24 µg per tube), most above the top standard so quoted by extrapolation. Set B metagenomic sites and mixes are ~1–2 ng/µL (≈ 0.1–0.15 µg per tube) — confirming the low-yield outcome and the plan to redo the metagenomic extraction with smaller per-tube biomass.